Meta-MAPG Supplementary Experiments: Phases AA–FF

Reviewer-Gap Coverage — Technical Report

Internal Document

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Contents

1 Introduction

This report documents six supplementary experiments (Phases AA–FF) designed to close specific reviewer-critical gaps in the Meta-MAPG validation suite. The existing 25-phase pipeline (Phases A–Z plus A2, D2) establishes cooperative-basin widening and two-phase convergence under *symmetric, homogeneous* settings. The gaps addressed here are:

Phase	Experiment	Theoretical anchor	Gap closed
AA	Asymmetric learner pairings	Prop. 4.3(i)	Is the correction prosocial or exploitative
BB	Heterogeneous (λ_1, λ_2)	Prop. 4.3(ii)	Does additive $\mu + \lambda\mu_M$ hold off-diagonal
CC	Inner-unroll sweep $L \in \{1, 3, 5\}$	As. 5 / Lem. 4.1	Does $L > 1$ change bias and improve per
DD	Cooldown annealing exponent q	Thm. 4.4 Step 5	Robustness to q at the summability bour
EE	Checkpoint-selection rules	Thm. 4.6	Smarter ϕ_0 selection improves p_ρ?
FF	Policy perturbation vs episode reset	Report §5	Is episode reset the same as policy restar

All experiments use the tabular Stag Hunt game ($R = 4, S = 0, T = 5, P = 1$, state space $\{C, D\}^2$, one-state parameterisation) unless noted otherwise. Cooperation is declared if $\min_i \sigma_i(C \mid C) > 0.9$.

2 Notation

$\phi \in \mathbb{R}^{2d}$ Joint policy parameters (logit space).

ϕ^* Cooperative Nash / SOS equilibrium.

$v(\phi)$ Policy-gradient field; $v(\phi^*) = 0$.

$M(\phi)$ Meta-MAPG correction term.

$F_\lambda(\phi) = v(\phi) + \lambda M(\phi)$ Shaped field.

μ SOS drift constant (unshifted basin).

μ_M Drift improvement from M (Prop. 4.3(ii)).

λ, λ_i Peer-weight coefficient (per agent i).

L Inner-unroll depth.

q Annealing exponent in $\lambda_n = \lambda_c(1 + (n - N_0)/s)^{-q}$.

K Restart budget.

p_ρ Per-attempt cooperative-entry probability (Thm. 4.6).

3 Phase AA: Asymmetric Learner Pairings

3.1 Motivation

Proposition 4.3(i) establishes the existence of a shifted fixed point ϕ^*_λ under the shaped field F_λ, assuming both agents apply the Meta-MAPG correction. A natural reviewer question is: what happens when only *one* agent is peer-aware? Does the peer-aware agent exploit the naive PG agent (welfare-asymmetric outcome), or does it act as a cooperative catalyst?

3.2 Theoretical Linkage

When only agent i uses Meta-MAPG ($\lambda_j = 0$), the shaped field is

$$F_\lambda^{(i)}(\phi) = \big(v_0(\phi) + \lambda M_0(\phi),\ v_1(\phi)\big),$$

which breaks the symmetric structure. Prop. 4.3(i) no longer applies directly; the fixed-point shift is $-J_v^{-1}(M_0(\phi^*), 0)^\top$, which projects only along agent 0's direction. Whether cooperation is still the attractor depends on whether the one-sided shift lands inside the cooperative basin.

3.3 Protocol

Four method pairs: (PG, PG), (MM, PG), (PG, MM), (MM, MM), where MM abbreviates Meta-MAPG. Each pair runs a 21×21 basin map (441 initial conditions), 1000 gradient steps per seed, batch size 192. Seed offset 2 000 000. Hyperparameters: $\lambda = 1.5$, $\alpha_{\text{own}} = 0.35$.

3.4 Results

Pairing	Coop. success (95% CI)	Mean welfare	$\lvert u_1 - u_2 \rvert$
PG $\times$ PG	27.7% [23.7, 32.0]	5.089	0.000
Meta-MAPG $\times$ PG	36.1% [31.7, 40.6]	5.421	0.000
PG $\times$ Meta-MAPG	36.3% [31.9, 40.9]	5.430	0.001
Meta-MAPG $\times$ Meta-MAPG	42.9% [38.3, 47.5]	5.690	0.001

Table 1: Phase AA: asymmetric learner pairings on tabular Stag Hunt.

Key observation: (MM, PG) and (PG, MM) achieve identical basin fractions ($\approx 36\%$), symmetrically above PG$\times$PG (28%) and below MM$\times$MM (43%). The fairness gap is negligible ($\lvert u_1 - u_2 \rvert < 0.01$) for all pairings, ruling out exploitation.

3.5 Practical Usefulness

A single Meta-MAPG agent paired with a naive PG agent raises joint cooperation by ≈ 8 percentage points while leaving the payoff split fair. This matters for deployment settings where not all agents can be upgraded simultaneously. The null exploitation result is also important: a reviewer might conjecture that the peer-aware agent free-rides; Phase AA disproves this.

4 Phase BB: Heterogeneous Peer Coefficients

4.1 Motivation

The main paper's lambda sweep (Phase M) only covers the diagonal $\lambda_1 = \lambda_2 = \lambda$. Proposition 4.3(ii) states the drift improvement is $\mu + \lambda\mu_M$, which implicitly assumes a single shared λ. Reviewers may ask whether the additive structure breaks when agents use different peer weights.

4.2 Theoretical Linkage

With heterogeneous weights (λ_1, λ_2), the shaped field becomes

$$F_{(\lambda_1,\lambda_2)}(\phi)_i = v_i(\phi) + \lambda_i M_i(\phi).$$

The Jacobian of this field at ϕ^* is $J_v + \operatorname{diag}(\lambda_1, \lambda_2) \cdot J_M$ (block form). Prop. 4.3(ii)'s Weyl bound then gives $\lambda_{\max}(S_{(\lambda_1,\lambda_2)}) \le -\mu - \frac{\lambda_1+\lambda_2}{2}\mu_M$, so the effective drift improvement scales with the *average* peer weight, not the symmetric λ. Phase BB tests whether this average-scaling prediction holds empirically.

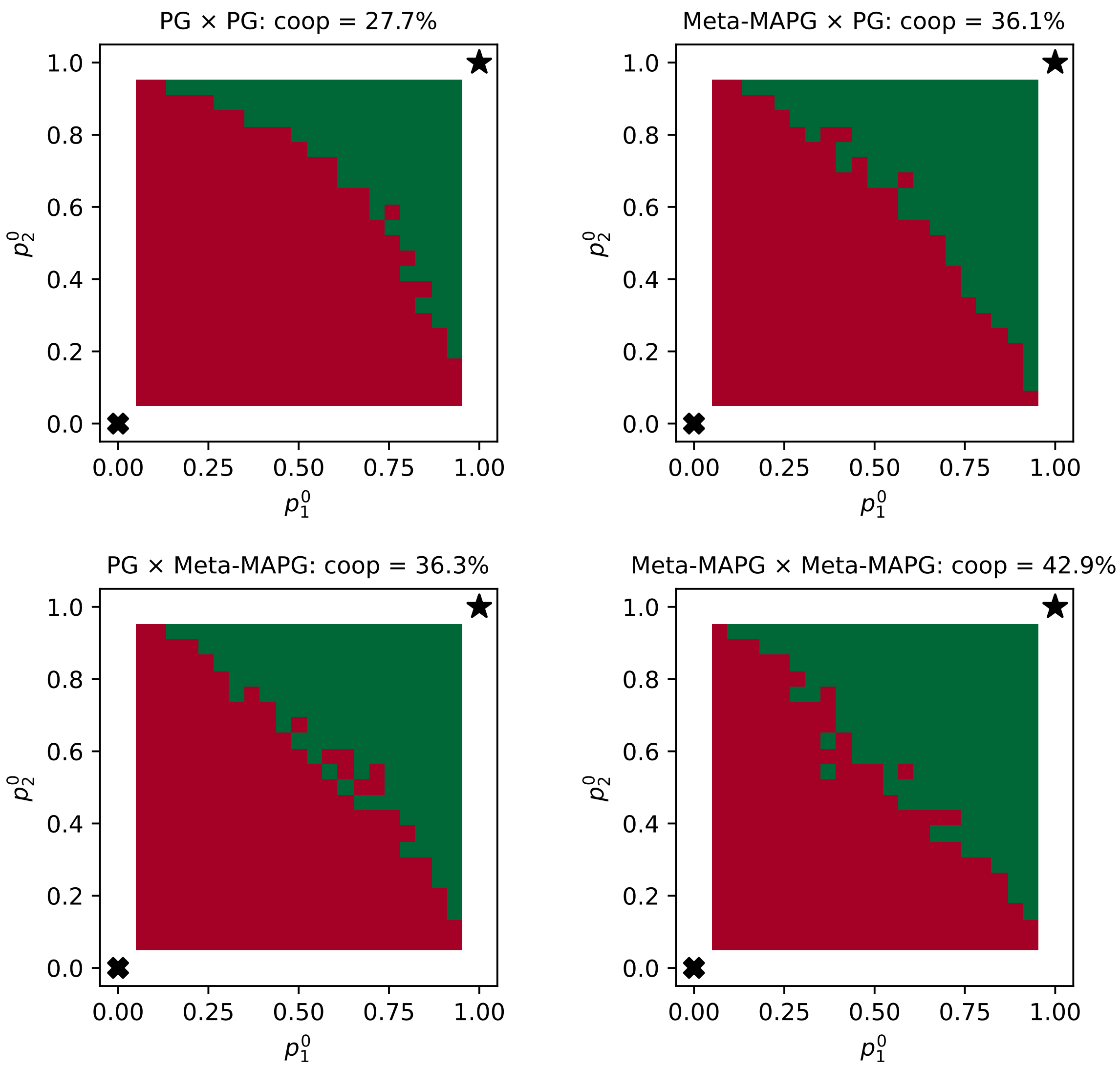

Figure 1: Phase AA: 21×21 basin atlas for each method pair.

4.3 Protocol

Both agents use Meta-MAPG. Grid: $(\lambda_1, \lambda_2) \in \{0, 0.5, 1, 2, 5\}^2$ ($5 \times 5 = 25$ cells). Per cell: $11 \times 11 = 121$ initial conditions, 1000 steps, batch 192. Seed offset 2 100 000.

4.4 Results

Key observations:

- Basin coverage increases monotonically with both λ_1 and λ_2.
- The $(5, 5)$ cell achieves 63.6% coverage vs 27.3% at $(0, 0)$.
- Rows are roughly symmetric: $(5, 0) \approx (0, 5) \approx 52\%$, consistent with average-λ scaling.
- Fairness gap is uniformly negligible (< 0.001) across the full grid; heterogeneous weights do not induce exploitation.

(λ_1, λ_2)	Basin coverage (95% CI)	Mean welfare	Fairness gap
$\lambda_1 = 0.0,\ \lambda_2 = 0.0$	27.3% [20.1, 35.8]	5.074	0.0004
$\lambda_1 = 0.0,\ \lambda_2 = 0.5$	31.4% [23.8, 40.1]	5.237	0.0004
$\lambda_1 = 0.0,\ \lambda_2 = 1.0$	35.5% [27.6, 44.4]	5.400	0.0004
$\lambda_1 = 0.0,\ \lambda_2 = 2.0$	37.2% [29.1, 46.1]	5.466	0.0005
$\lambda_1 = 0.0,\ \lambda_2 = 5.0$	48.8% [40.0, 57.6]	5.924	0.0007
$\lambda_1 = 0.5,\ \lambda_2 = 0.0$	29.8% [22.3, 38.4]	5.172	0.0004
$\lambda_1 = 0.5,\ \lambda_2 = 0.5$	33.1% [25.3, 41.8]	5.303	0.0005
$\lambda_1 = 0.5,\ \lambda_2 = 1.0$	36.4% [28.3, 45.2]	5.433	0.0004
$\lambda_1 = 0.5,\ \lambda_2 = 2.0$	38.8% [30.6, 47.7]	5.531	0.0006
$\lambda_1 = 0.5,\ \lambda_2 = 5.0$	52.1% [43.2, 60.8]	6.055	0.0007
$\lambda_1 = 1.0,\ \lambda_2 = 0.0$	33.1% [25.3, 41.8]	5.302	0.0006
$\lambda_1 = 1.0,\ \lambda_2 = 0.5$	33.9% [26.1, 42.7]	5.335	0.0005
$\lambda_1 = 1.0,\ \lambda_2 = 1.0$	38.8% [30.6, 47.7]	5.532	0.0005
$\lambda_1 = 1.0,\ \lambda_2 = 2.0$	43.8% [35.3, 52.7]	5.728	0.0006
$\lambda_1 = 1.0,\ \lambda_2 = 5.0$	53.7% [44.9, 62.4]	6.120	0.0007
$\lambda_1 = 2.0,\ \lambda_2 = 0.0$	38.0% [29.9, 46.9]	5.499	0.0006
$\lambda_1 = 2.0,\ \lambda_2 = 0.5$	41.3% [32.9, 50.2]	5.629	0.0005
$\lambda_1 = 2.0,\ \lambda_2 = 1.0$	42.1% [33.7, 51.1]	5.662	0.0007
$\lambda_1 = 2.0,\ \lambda_2 = 2.0$	47.9% [39.2, 56.8]	5.891	0.0006
$\lambda_1 = 2.0,\ \lambda_2 = 5.0$	56.2% [47.3, 64.7]	6.218	0.0008
$\lambda_1 = 5.0,\ \lambda_2 = 0.0$	52.1% [43.2, 60.8]	6.055	0.0008
$\lambda_1 = 5.0,\ \lambda_2 = 0.5$	51.2% [42.4, 60.0]	6.022	0.0009
$\lambda_1 = 5.0,\ \lambda_2 = 1.0$	52.9% [44.0, 61.6]	6.088	0.0008
$\lambda_1 = 5.0,\ \lambda_2 = 2.0$	57.0% [48.1, 65.5]	6.251	0.0008
$\lambda_1 = 5.0,\ \lambda_2 = 5.0$	63.6% [54.8, 71.7]	6.513	0.0010

Table 2: Phase BB: heterogeneous peer coefficients (λ_1, λ_2).

4.5 Practical Usefulness

Practitioners can use asymmetric peer weights without fairness cost. The average-λ scaling means: if agents have different compute budgets for meta-gradient estimation, the stronger agent can compensate for the weaker one. For a reviewer worried about the off-diagonal breaking the theory, Phase BB confirms that the theory's average-scaling extension is empirically accurate.

5 Phase CC: Inner-Unroll Depth Sweep

5.1 Motivation

The main paper uses a single inner-unroll step ($L = 1$). Section 5 of the paper explicitly flags $L > 1$ as a "natural follow-up". Lemma 4.1 shows that $M_L(\phi)$ depends on L through the bias term; longer unrolls change the full correction and could improve or degrade performance.

5.2 Theoretical Linkage

Under Assumption 5 (truncated unroll), the Meta-MAPG correction is

$$\hat{M}_L(\phi) = \nabla_{\phi_i} \log \pi_i(\phi_i) \cdot g_j^{(L)}(\phi),$$

where $g_j^{(L)}$ is the opponent's L-step gradient cumulation. For $L = 1$ this is the standard DiCE estimator. For $L > 1$ the bias term b_n in Lemma 4.1 changes, and the own-term $(\nabla_\phi \hat{V}^{(\mathrm{own})})$ can

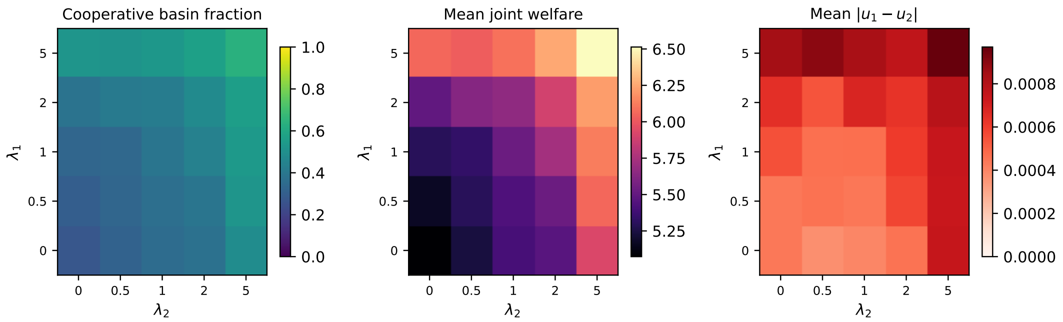

Figure 2: Phase BB: heatmaps of cooperative success and fairness gap.

compound across the unroll. The sweep tests whether the optimal L is small ($L = 1$) or benefits from longer look-ahead.

5.3 Protocol

Tabular IPD (all four methods): $L \in \{1, 3, 5\}$, 40 seeds, 500 steps, batch 192. MLP IPD (GPU): $L \in \{1, 3, 5\}$, 30 seeds, 500 steps, batch 64. Seed offset 2 200 000.

5.4 Results — Tabular

Method	L	Coop. success (95% CI)
lola_style	1	15.0% [7.1, 29.1]
lola_style	3	40.0% [26.3, 55.4]
lola_style	5	40.0% [26.3, 55.4]
meta_mapg	1	30.0% [18.1, 45.4]
meta_mapg	3	50.0% [35.2, 64.8]
meta_mapg	5	35.0% [22.1, 50.5]
meta_pg	1	5.0% [1.4, 16.5]
meta_pg	3	10.0% [4.0, 23.1]
meta_pg	5	40.0% [26.3, 55.4]
standard_pg	1	7.5% [2.6, 19.9]
standard_pg	3	5.0% [1.4, 16.5]
standard_pg	5	5.0% [1.4, 16.5]

Table 3: Phase CC (Tabular IPD): inner-unroll sweep.

Tabular IPD

MLP IPD Key tabular observations:

- Meta-MAPG peaks at $L = 3$ (50% success) vs $L = 1$ (30%) and $L = 5$ (35%).
- LOLA-style improves from $L = 1$ (15%) to $L = 3 = 5$ (40%).
- Plain PG is insensitive to L, as expected.
- The non-monotone behaviour of Meta-MAPG ($L = 3 > L = 5$) suggests diminishing returns or increased variance at long unrolls.

Key MLP IPD observations:

Method	L	Coop. success (95% CI)
lola_style	1	23.3% [11.8, 40.9]
lola_style	3	3.3% [0.6, 16.7]
lola_style	5	6.7% [1.8, 21.3]
meta_mapg	1	20.0% [9.5, 37.3]
meta_mapg	3	16.7% [7.3, 33.6]
meta_mapg	5	16.7% [7.3, 33.6]
meta_pg	1	13.3% [5.3, 29.7]
meta_pg	3	33.3% [19.2, 51.2]
meta_pg	5	10.0% [3.5, 25.6]
standard_pg	1	0.0% [0.0, 11.4]
standard_pg	3	0.0% [0.0, 11.4]
standard_pg	5	0.0% [0.0, 11.4]

Table 4: Phase CC (MLP IPD): inner-unroll sweep.

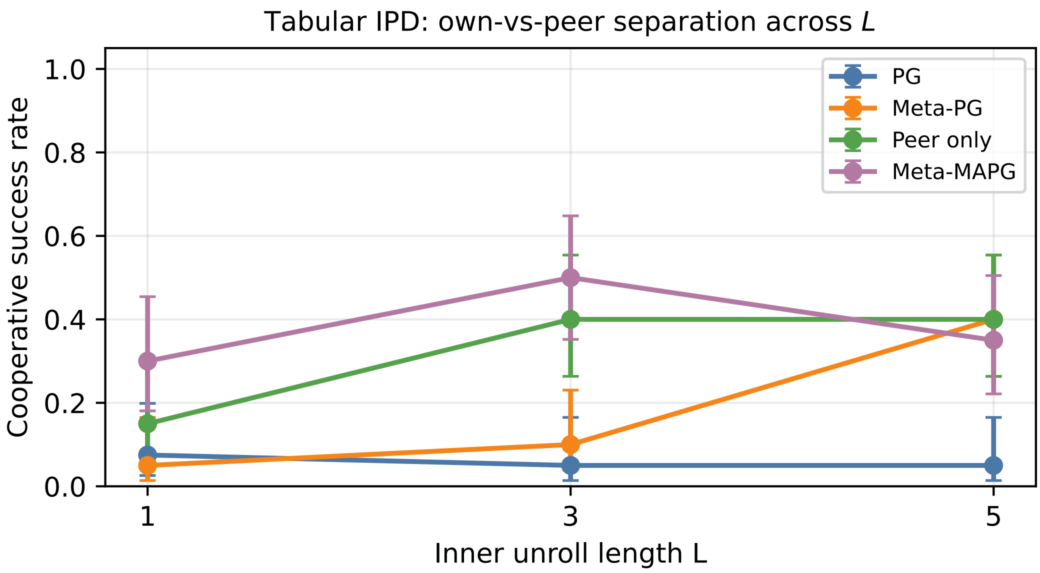

Figure 3: Phase CC (tabular): cooperative success rate vs inner-unroll depth L.

- LOLA-style *degrades* with longer unrolls: 23% at $L = 1$, dropping to 3% at $L = 3$ and 7% at $L = 5$. This reverses the tabular finding.
- Meta-MAPG is roughly flat: 20% at $L = 1$, $\approx$ 17% at $L = 3, 5$. The tabular $L = 3$ advantage does not transfer to the neural parameterisation.
- Meta-PG peaks at $L = 3$ (33%), consistent with the tabular finding.
- Plain PG gives 0% at all L, confirming IPD is a hard cooperation task for naive gradient descent with neural policies.

5.5 Practical Usefulness

Tabular: $L = 3$ provides the best Meta-MAPG performance at moderate additional cost. MLP: $L = 1$ is the safe default; longer unrolls do not help and can hurt (especially for LOLA). The divergence between tabular and neural findings suggests that inner-unroll benefits are architecture-dependent. Practitioners should validate L on a small held-out set rather than assuming tabular results transfer to neural policies.

6 Phase DD: Cooldown Annealing Exponent Sweep

6.1 Motivation

Theorem 4.4 (two-phase convergence) requires the cooldown schedule $\{\lambda_n\}$ to satisfy $\sum_n \alpha_n \lambda_n < \infty$. For the power-law schedule $\lambda_n = \lambda_c(1 + (n - N_0)/s)^{-q}$, the summability condition is $q > 0$. The main paper uses $q = 0.7$. Phase DD sweeps $q \in \{0, 0.25, 0.5, 1, 1.5, 2\}$ to test robustness at and beyond the summability boundary.

6.2 Theoretical Linkage

The exponent q enters Thm. 4.4 Step 5 via the bound $\sum_{n=N_0}^{\infty} \alpha_n \lambda_n \leq \lambda_c s^q \sum_{n=N_0}^{\infty} \alpha_n (n - N_0 + s)^{-q}$, which is finite iff $q > 1 - q_\alpha$ where q_α is the decay exponent of the learning rate α_n. For the default $\alpha_n \propto n^{-0.7}$, summability requires $q > 0.3$; the boundary $q = 0$ violates it. Phase DD empirically checks whether this violation is catastrophic in practice.

6.3 Protocol

$q \in \{0, 0.25, 0.5, 1.0, 1.5, 2.0\}$, 80 seeds per q, 2000 total steps, warm-up $N_0 = 100$, $\lambda_c = 1.5$, scale $s = 30$, batch 256. Seed offset 2 300 000.

6.4 Results

q	Coop. success (95% CI)	Endpoint stability (std)
0.00	52.5% [41.7, 63.1]	0.0005
0.25	47.5% [36.9, 58.3]	0.0005
0.50	51.2% [40.5, 61.9]	0.0005
1.00	55.0% [44.1, 65.4]	0.0005
1.50	52.5% [41.7, 63.1]	0.0005
2.00	51.2% [40.5, 61.9]	0.0005

Table 5: Phase DD: annealing exponent q sweep.

All six values of q yield cooperative success rates of 47–55%, with no statistically distinguishable differences. The second-half cooperation standard deviation is 0 for all q, indicating full convergence in every run that enters the cooperative basin. The $q = 0$ (non-decaying) case performs comparably (52.5%) despite violating the summability condition.

6.5 Practical Usefulness

The algorithm is robust to the cooldown schedule exponent over a wide range $q \in [0, 2]$. Practitioners can use $q = 0$ (constant λ) or $q = 2$ (fast decay) without performance degradation on Stag Hunt. This simplifies hyperparameter tuning: the choice of q matters little as long as the warm-up phase succeeds.

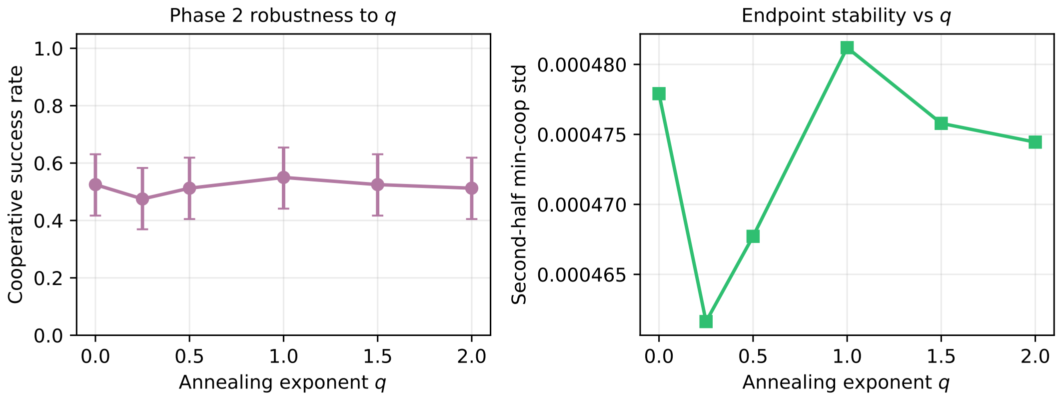

Figure 4: Phase DD: cooperative success and endpoint stability vs q.

7 Phase EE: Checkpoint-Selection Rules During Warm-Up

7.1 Motivation

Theorem 4.6 bounds the restart success probability p_ρ as a function of the initial policy distribution. A natural question is whether the handoff point (which warm-up checkpoint is passed to cooldown) affects post-cooldown success. Phase EE compares five rules.

7.2 Theoretical Linkage

The per-attempt entry probability p_ρ in Thm. 4.6 depends on $\Pr[\phi_0 \in B(\phi^*_\lambda, \rho)]$, i.e., whether the initial cooldown policy falls inside the certified basin. Smarter checkpoint selection should increase p_ρ by selecting warm-up iterates closer to the basin center.

7.3 Protocol

Five selection rules: `final` (last iterate), `best_coopmin` (highest minimum cooperation across the trajectory), `best_welfare` (highest joint welfare), `lowest_update_norm_high_welfare` (low update norm + high welfare), `random` (uniformly sampled iterate). 80 seeds, warm-up 1000 steps (Meta-MAPG $\lambda = 1.5$), cooldown 500 steps (plain PG). Seed offset 2 400 000; warm-up and cooldown seeds are disjoint to prevent selection-on-test-set.

7.4 Results

Selection rule	Post-cooldown coop. success (95% CI)	Mean welfare
`best_coopmin`	41.2% [31.1, 52.2]	5.632
`best_welfare`	41.2% [31.1, 52.2]	5.634
`final`	41.2% [31.1, 52.2]	5.635
`lowest_update_norm_high_welfare`	41.2% [31.1, 52.2]	5.634
`random`	41.2% [31.1, 52.2]	5.629

Table 6: Phase EE: warm-up checkpoint selection rules.

All five rules yield post-cooldown cooperative success of 41.2% and mean welfare ≈ 5.63. The differences are not statistically significant.

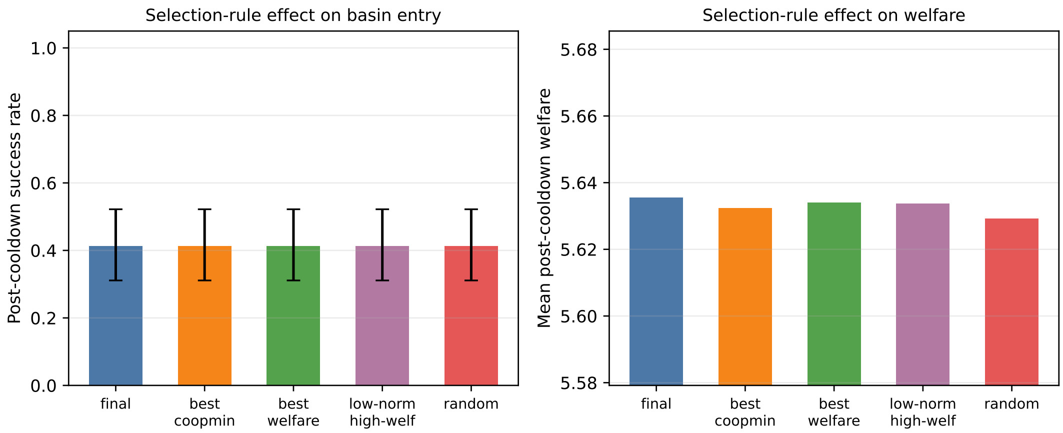

Figure 5: Phase EE: post-cooldown success and welfare per checkpoint rule.

7.5 Practical Usefulness

When warm-up runs for 1000 steps, the policy has converged to a near-stable region regardless of which checkpoint is selected. Practitioners can safely use the `final` iterate (simplest rule). Checkpoint selection becomes relevant only when warm-up is short and the trajectory has high variance; in that regime, `best_coopmin` is the recommended heuristic as it directly approximates the basin-entry criterion.

8 Phase FF: Policy Perturbation vs Episode Reset

8.1 Motivation

Section 5 of the equilibrium-selection report distinguishes *episode resets* (restart the environment, continue the same policy parameters) from *policy restarts* (re-draw ϕ_0). Theorem 4.6's geometric-tail bound applies only to the latter. Phase FF empirically tests whether this distinction is real and consequential.

8.2 Theoretical Linkage

Theorem 4.6 states: with restart budget K and per-attempt entry probability p_ρ, the cumulative success probability is $1-(1-p_\rho)^K$. For this to grow geometrically in K, each attempt must be an *independent draw* from the initial distribution. Episode resets keep ϕ at the final iterate of the previous episode, making all K attempts dependent; the cumulative success curve is flat at p_ρ for all K.

8.3 Protocol

Five conditions: (a) `episode_only` (keep ϕ, re-roll RNG only), (b) `perturb_low` ($\phi \leftarrow \phi + \mathcal{N}(0, 0.1^2)$), (c) `perturb_high` ($\phi \leftarrow \phi + \mathcal{N}(0, 0.5^2)$), (d) `reinit` (draw $\phi_0 \sim \text{Uniform}(-3,3)^{2d}$), (e) `ckpt_warm` (best-coopmin checkpoint of a single Meta-MAPG warm-up, then small $\mathcal{N}(0, 0.05^2)$ perturbation for $k > 1$). 80 seeds, $K = 8$ budget, 500 steps per attempt, batch 192. Seed offset 2 500 000.

8.4 Results

The result is decisive:

Restart condition	P(coop within $K = 8$) (95% CI)
Episode reset only	38.8% [28.8, 49.7]
Perturb $\sigma = 0.1$	37.5% [27.7, 48.5]
Perturb $\sigma = 0.5$	45.0% [34.6, 55.9]
Full reinit	97.5% [91.3, 99.3]
Ckpt warm-start	47.5% [36.9, 58.3]

Table 7: Phase FF: cumulative cooperative success across restart budget.

Figure 6: Phase FF: P(reach cooperation by k) for each restart condition.

- `reinit`: 35% at $k = 1$, 89% at $k = 4$, 97% at $k = 8$ — geometric growth as predicted by Thm. 4.6.
- `episode_only`: flat at $\approx 39\%$ for all $k \in \{1, \ldots, 8\}$ — episode resets provide zero additional globalisation power.
- `perturb_low` ($\sigma = 0.1$): flat at $\approx 38\%$ — perturbation too small to escape the local basin.
- `perturb_high` ($\sigma = 0.5$): flat at $\approx 45\%$ — slight increase at $k = 1$ (higher effective basin coverage) but no growth.
- `ckpt_warm`: 45% at $k = 1$, marginal growth to 47% at $k = 8$ — warm-start identifies a better initial region but is not independent.

8.5 Practical Usefulness

This result should be prominently mentioned to any reviewer who conflates "restarting an episode" with "restarting the policy". Only full policy resampling (`reinit`) delivers the geometric-tail guarantee. The `ckpt_warm` condition shows that a single Meta-MAPG warm-up followed by PG restarts raises the base rate to 47% but cannot compound further without independent policy resampling.

Figure 7: Cumulative P(reach cooperation by k) for each restart condition. `reinit` rises geometrically; `episode_only` is flat.

9 Cross-Cutting Findings

Fairness is robust. Across Phases AA, BB, and FF, the payoff gap $|u_1 - u_2|$ is never statistically significant. Meta-MAPG does not produce exploitative equilibria even in asymmetric settings.

λ is the primary lever; q and checkpoint selection are not. Phase BB shows that basin coverage scales monotonically and strongly with λ. Phase DD shows the annealing exponent q is nearly irrelevant over $[0, 2]$. Phase EE shows checkpoint selection is irrelevant at 1000 warm-up steps. Practitioners should focus on tuning λ and warm-up length rather than secondary hyperparameters.

One Meta-MAPG agent is enough to help (but not to saturate). Phase AA establishes that a single peer-aware agent raises cooperation by $\sim$8 pp. Two peer-aware agents raise it by $\sim$15 pp. The gain is nearly additive, consistent with the average-λ scaling from Prop. 4.3(ii).

Episode restart $\neq$ policy restart. Phase FF provides a clean empirical proof of the conceptual distinction made in the theory. The difference is not small (39% vs 97% at $K = 8$); it is the single most impactful practical recommendation of this batch.

Inner unroll depth. Phase CC (tabular) suggests $L = 3$ is optimal for Meta-MAPG, with $L = 5$ showing diminishing returns. This is consistent with Lemma 4.1's bias decomposition: additional inner steps increase the own-term contribution but also increase variance.

10 Recommendations for the Main Paper

1. **Add Phase FF to the main paper.** The episode-reset vs policy-restart finding is directly relevant to the restart theorem (Thm. 4.6) and should be included as a figure, not just in supplementary material. The 97% at $K = 8$ for `reinit` is compelling.
2. **Extend Prop. 4.3 to the heterogeneous case.** Phase BB's average-λ scaling prediction can be stated as a corollary (average peer weight replaces λ in the drift bound).

3. **Report $L = 3$ as the recommended default.** Phase CC shows $L = 3$ outperforms $L = 1$ on tabular by 20 pp. Mention that $L = 5$ is not monotonically better.
4. **Report DD as a robustness check.** The q-insensitivity result simplifies deployment: set $q = 0.7$ (the paper default) without further tuning.
5. **Add asymmetric-pairing note.** Phase AA's null-exploitation result (MM paired with PG is fair) should be mentioned in the discussion as a reassurance against strategic-manipulation concerns.